Q.25 What is an inverter? Write any three applications.

Q.26 What is a chopper? Explain its different types.

Q.27 Explain the working of a dual converter in detail.

Q.28 Explain working of a single -phase cycloconverter using centre tapped transformer.

Q.29 What is electric drive? Explain it with the help of block diagram of an electric drive system?

Q.30 Explain the working of single phase half wave converter drive with waveforms.

Q.31 What is UPS? Explain different types of UPS?

Q.32 Discuss the basic idea of heat sink selection for SCRs. Why is it important?

Q.33 What are the specifications and ratings of SCR? Explain in detail.

Q.34 Discuss the concept of high voltage DC transmission. Why is it preferred in certain applications?

Q.35 Write short note on “thyristor protection”.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the construction and working principle of TRIAC and DIAC with their V-I characteristics.

Q.37 Explain the working principle and characteristics of UJT. How is it used as a relaxation oscillator?

Q.38 Explain with the help of suitbale diagram of an automatic battery charger.

Roll No. ....

4th Sem / Eltx

Subject:- Power Electronics

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The primary function of an SCR in a power electronics circuit is as:

a) Voltage control b) Current control

c) Power amplification d) Switching devices

Q.2 Following is a type of inverter:

a) Step-up inverter b) Step-down inverter

c) Both a and b d) None of these

Q.3 The purpose of heat sink in an SCR circuit is:

a) To increase voltage

b) To stabilize the power output

c) To dissipates heat and prevents damage

d) To regulate current

Q.4 The primary purpose of cycloconverter is as:

a) Converting AC to DC

b) Step-up conversion

c) Frequency conversion

d) Inverter operation

Q.5 In an Off-line UPS system, power is supplied to the load:

a) Only from the battery
b) From the inverter
c) Directly from the mains, with battery backup in case of power failure
d) None of above

Q.6 UJT is widely used as:

a) Oscillator b) Pulse circuit
c) Both (a) and (b) above
d) None of above

Q.7 A single phase full wave fully controlled bridge rectifier uses:

a) 4 SCRs b) 6 SCRs
c) 2 SCRs d) 3 SCRs

Q.8 A dual converter consists of two bridges out of which one works as :

a) A rectifier the other as oscillator
b) A inverter the other as chopper
c) A inverter the other as frequency changer
d) A rectifier the other as inverter

Q.9 A class-E chopper can operate in:

a) 1st and 2nd quadrant b) 2nd and 3rd quadrant
c) 1st and 4th quadrant d) All the four quadrant

Q.10 To turn off an SCR,it is necessary to reduce its current to less than:

a) Trigger current b) Holding current
c) Break over current d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 UJT stands for: _________

Q.12 The maximum efficiency of a half-wave controlled rectifier is:________

Q.13 An AC gate triggering is preferred over DC gate triggering. (True/False)

Q.14 A DIAC is a unidirectional device.(True/False)

Q.15 The output voltage of a controlled rectifier is maximum when firing angle is_______.

Q.16 The duty cycle of a chopper is expressed as_______

Q.17 SMF batteries are not costlier than lead acid batteries.(True/False)

Q.18 Armature control method is used in dc motors to get speed______ normal speed.

Q.19 Step up chopper does not require forced commutation. (True/False)

Q.20 LASCR stand for__________.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the working of SCR using its layer diagram.

Q.22 Differentiate between holding current and latching currrent of a thyristor.

Q.23 Differentiate between natural and forced commutation.

Q.24 Explain the working of single phase full wave half controlled bridge rectifier.